Unit 8, Lesson 3: Rewriting Quadratic Equations

Benchmarks

MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros, and interpret them in terms of a real-world context.

MA.912.AR.3.7 Given a table, equation, or written description of a quadratic function, graph that function and determine and interpret its key features.

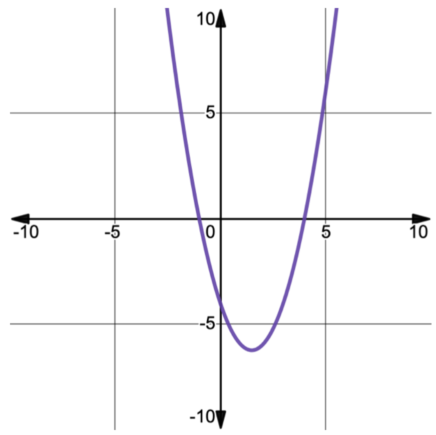
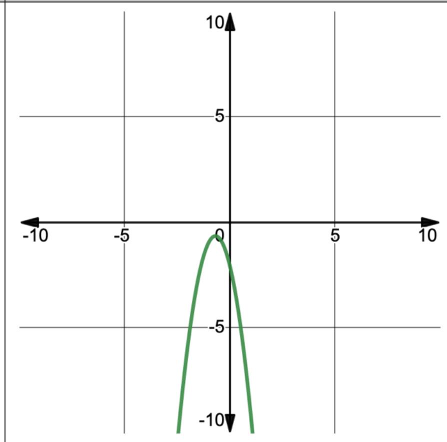
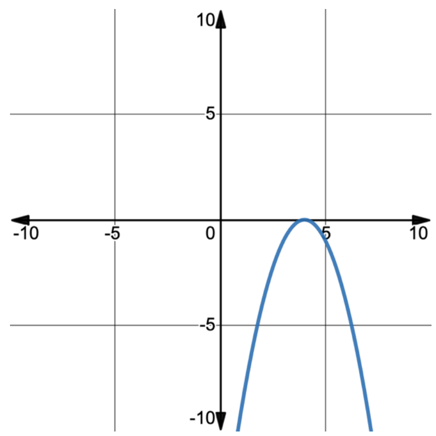
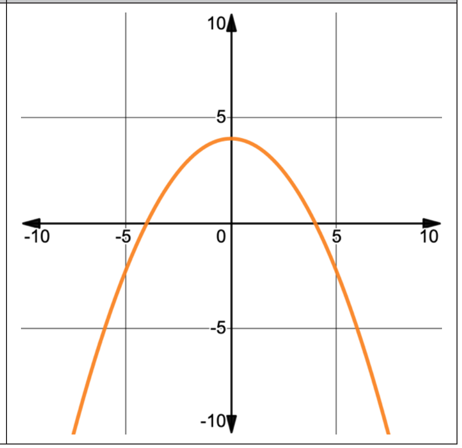
MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

Learning Targets

- ☐ I can rewrite a quadratic equation from factored or vertex form to standard form.
- ☐ I can rewrite a quadratic equation from standard form to factored form.
- ☐ I can determine key features of a quadratic function from a variety of forms.

8.3.1 Warm-Up: Which One Doesn't Belong?

1. Four graphs are shown.

Graph A	Graph B
	
Graph C	**Graph D**
	

a. Which graph doesn't belong?

b. Justify your reasoning using key features of the graphs.

8.3.2 Guided Instruction: Forms of Quadratic Equations

Quadratic equations can be represented in three forms: vertex form, factored (x-intercept) form, and standard form.

Forms of Quadratic Equations

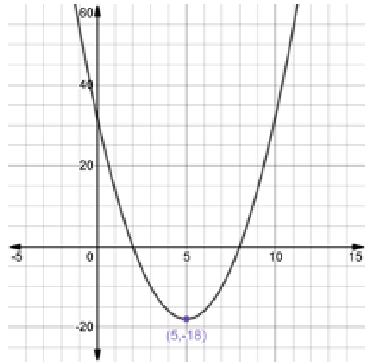
Vertex form of a quadratic equation is represented as $y = a(x - h)^2 + k$, where (h, k) is the vertex.

Factored form of a quadratic equation is represented as $y = a(x - p)(x - q)$, where $(p, 0)$ and $(q, 0)$ are the x-intercepts.

Standard form of a quadratic equation is represented as $y = ax^2 + bx + c$, where $(0, c)$ is the y-intercept.

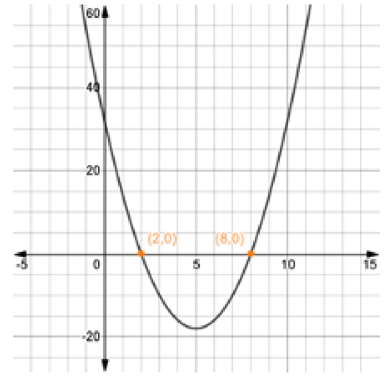
Each form highlights key features that can be used to assist in graphing quadratic equations.

<table>
<tr><th>Form of the Equation</th><th>Graph of the Equation</th></tr>
<tr><td>Vertex form of a quadratic equation $y = a(x - h)^2 + k$

Example shown:
$y = 2(x - 5)^2 - 18$
Vertex $(5, -18)$
Axis of symmetry $x = 5$
Parabola opens up</td><td></td></tr>
<tr><th colspan="2">Key Features</th></tr>
<tr><td colspan="2">

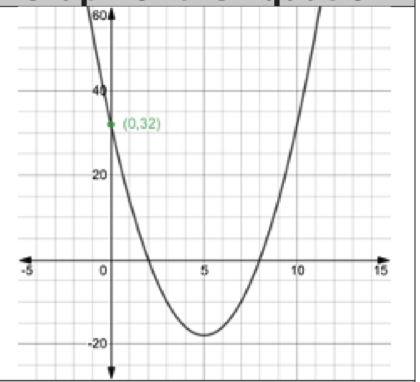
- Highlights the vertex of a parabola (h, k) and the axis of symmetry $x = h$.
- Highlights whether the graph opens up or down.
  - If $a > 0$ (positive), the graph opens up.
  - If $a < 0$ (negative), the graph opens down.

</td></tr>
</table>

Form of the Equation	Graph of the Equation
Factored form of a quadratic equation $y = a(x-p)(x-q)$ Example shown: $y = 2(x-8)(x-2)$ x-intercepts $(8,0)(2,0)$ Parabola opens up	

Key Features

- Highlights the intercepts of a parabola $(p,0)(q,0)$.
- Highlights whether the graph opens up or down.

Form of the Equation	Graph of the Equation
Standard form of a quadratic equation $y = ax^2 + bx + c$ Example shown: $y = 2x^2 - 20x + 32$ y-intercept $(0,32)$ Parabola opens up	

Key Features

- Highlights the y-intercept $(0,c)$.
- Easily shows the coefficients for a, b and c, which can be used to determine other key features of quadratics.
- Highlights whether the graph opens up or down.

1. What do you notice about the graphs of the three different quadratic equations?

2. Complete the table for each equation.

Equation	Form of the Equation	Key Feature(s) Revealed	Value(s) of Key Features
$y = (x-8)(x+13)$	○ vertex form ○ factored form ○ standard form		
$y = -\frac{1}{2}x^2 - 2x - 2$	○ vertex form ○ factored form ○ standard form		
$y = (x+6)^2 - 4$	○ vertex form ○ factored form ○ standard form		
$y = -x^2 - 12x + 36$	○ vertex form ○ factored form ○ standard form		
$y = 3(x+4)^2$	○ vertex form ○ factored form ○ standard form		

8.3.3 Collaboration: Rewriting Quadratic Equations - Part 1

1. Rewrite the quadratic equation in standard form.

$$y = (x - 3)(x + 1)$$

2. Rewrite the quadratic equation in standard form.

$$y = (x - 1)^2 - 4$$

3. With your partner, compare the two standard form equations and write what you observe.

4. Discuss with your partner if any key features could help determine if a quadratic in vertex form and a quadratic equation in factored form are equivalent. Write a conjecture.

8.3.4 Guided Instruction: Rewriting Quadratic Equations - Part 2

1. Rewrite the quadratic equation $y = -x^2 - 4x + 5$ in factored form and identify the x-intercepts.

2. Liam wants to jump over 15 chairs on his dirt bike. To calculate where the landing ramp must be placed, in meters, he uses the quadratic equation $y = -3x^2 + 21x$. Rewrite the equation in factored form and identify the x-intercepts.

3. A ball is thrown straight up from 2 meters above the ground with a velocity of 7 meters/second. The height of the ball is represented by the equation $y = -4x^2 + 7x + 2$, where x is time in seconds.

   a. Rewrite the equation in factored form.

b. Discuss with your partner what key features can be determined from each of the equations.

Equation	Standard Form $y = -4x^2 + 7x + 2$	Factored Form $y = \_(\_\_\_\_\_\_)(\_\_\_\_\_\_)$
Key Features Revealed		

c. Identify the x-intercepts.

d. Discuss with your partner what the x-intercepts mean in terms of the given context. Complete the sentence.

Because x is
○ time, in seconds,
○ height, in meters,
the

x-intercepts give the
○ time
○ height
when the

ball will
○ reach the ground.
○ be at its greatest height.

8.3.5 Guided Practice: Rewriting Quadratic Equations and Key Features

1. For each given equation, rewrite it into the specified form. Then, use both equations to determine key features of the quadratic. Show your work.
It may not be possible to determine all key features from the given form and the rewritten form.

Factored Form	Standard Form
$y = (x + 7)(x - 6)$	

Key Features			
x-intercept		increasing	
x-intercept		decreasing	
y-intercept		positive	
vertex		negative	
domain		opens up/down	
range		axis of symmetry	
left end behavior	$x \to -\infty$, $y \to$ _____	right end behavior	$x \to \infty$, $y \to$ _____

Factored Form	Standard Form
	$y = -x^2 - 13x + 30$

Key Features			
x-intercept		increasing	
x-intercept		decreasing	
y-intercept		positive	
vertex		negative	
domain		opens up/down	
range		axis of symmetry	
left end behavior	$x \to -\infty$, $y \to$ _____	right end behavior	$x \to \infty$, $y \to$ _____

Vertex Form	Standard Form
$y = -(x-3)^2 + 1$	

Key Features			
x-intercept		increasing	
x-intercept		decreasing	
y-intercept		positive	
vertex		negative	
domain		opens up/down	
range		axis of symmetry	
left end behavior	$x \to -\infty$, $y \to$ _____	right end behavior	$x \to \infty$, $y \to$ _____

Standard Form	Factored Form
$y = 2x^2 + 14x + 12$	

Key Features			
x-intercept		increasing	
x-intercept		decreasing	
y-intercept		positive	
vertex		negative	
domain		opens up/down	
range		axis of symmetry	
left end behavior	$x \to -\infty$, $y \to$ _____	right end behavior	$x \to \infty$, $y \to$ _____

8.3.6 Wrap-Up: Cool Down

1. Write a note to a friend who missed class summarizing today's lesson.